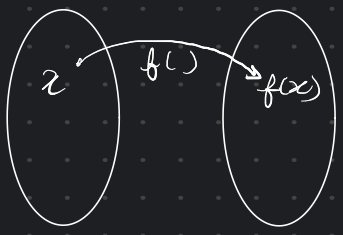
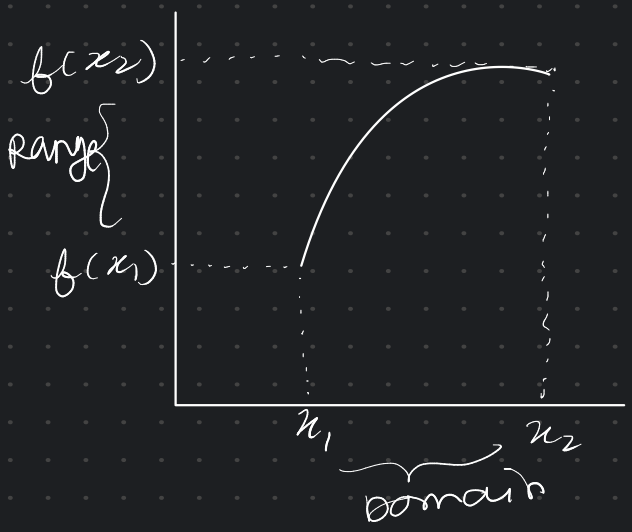
॥ श्रीगणेशाय नमः॥

Functions:

A function is a rule that assigns to each element $x \in D$ (domain) in a set to exactly one element called $f(x)$ in set R (range).





Examples on domain and range:

$f(x) = x$; $D \in \mathbb{R}$, $R \in \mathbb{R}$

$f(x) = x^2$; $D \in \mathbb{R}$ $R \in [0, \infty)$

$f(x) = \sqrt{x}$; $D \in [0, \infty)$, $R \in [0, \infty)$

Q. $f(x) = 2x^2 - 5x + 1$. $\frac{f(a+h) - f(a)}{h} = ?$

$$\rightarrow = \frac{2(a+h)^2 - 5(a+h) + 1 - 2a^2 - 5a - 1}{h}$$

$$= \frac{2a^2 + 4ah + 2h^2 - 5a - 5h + 1 - 2a^2 - 5a - 1}{h}$$

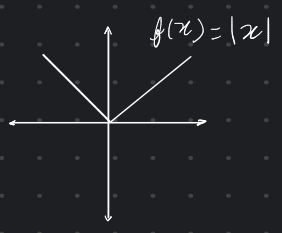
$$= \frac{4ah + 2h^2 - 5h}{h} = 4a + 2h - 5$$

$f(x) = \frac{1}{x}$ $D \in \mathbb{R} - \{0\}$, $R \in \mathbb{R} - \{0\}$

$f(x) = \sqrt{x+2}$; $x \in [-2, \infty)$, $R \in [0, \infty)$

$f(x) = \frac{1}{x^2 - x}$ $D \in \mathbb{R} - \{0, 1\}$, $R \in \mathbb{R} - \{0\}$

$$f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



$D \in \mathbb{R}$; $R \in [0, \infty)$

Even Functions:
Any function that satisfies f(-x) = f(x)

Odd Functions:
Any function that satisties f(-x) = -f(x)

e.g. $f(x) = x^2$
→ $f(-x) = (-x)^2 = x^2 = f(x)$ ⇒ even fⁿ

e.g. $f(x) = x^3$
→ $f(-x) = -x^3$ ⇒ odd fⁿ

e.g. $f(x) = |x|$
→ $f(-x) = |-x| = |x|$ ⇒ even

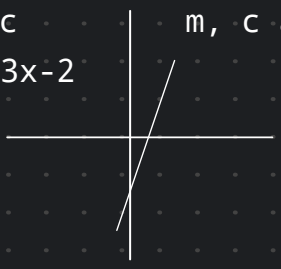
Q. $h(x) = 2x - x^2$
→ $h(-x) = -2x - x^2$ ⇒ neither even nor odd.

Increasing and decreasing functions:
- increasing if in an interval I, f(x1) < f(x2) if x1 < x2
- decreasing if in an interval I, f(x1) > f(x2) if x1 < x2

Standard Functions

Linear:
y = f(x) = mx+c m, c are constants

say y = 3x-2



Polynomial:

$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$

n → non-ve int

↳ of degree n if $a_n \neq 0$

eg. $p(x) = mx + c$

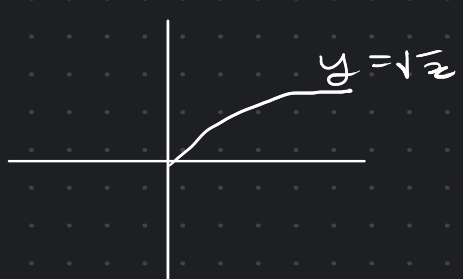
Power:

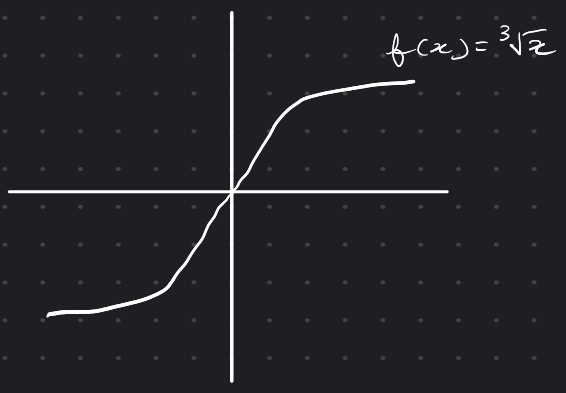
$f(x) = x^a$

① $a = n$, n is +ve int ⇒ polynomial func?

② $a = \frac{1}{n}$,

$f(x) = x^{1/n} = \sqrt[n]{x}$



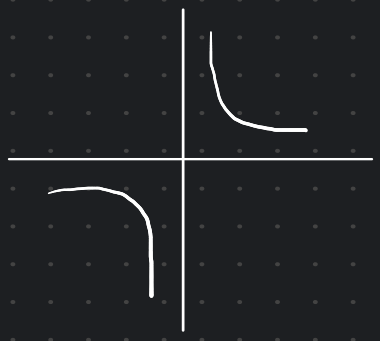


③ $a = -1$

$f(x) = x^{-1} = \frac{1}{x}$

$y = \frac{1}{x}$

$xy = 1$



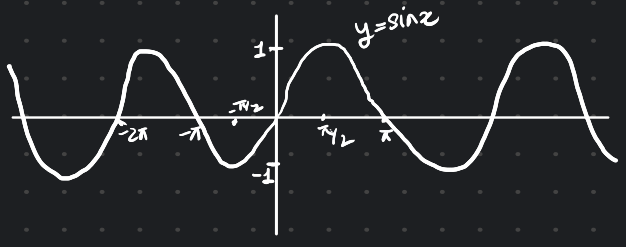
Rational Functions:
ratio of 2 polynomial functions

$$f(x) = \frac{p(x)}{q(x)} \qquad q(x) \neq 0 \qquad D = \{x \mid q(x) \neq 0\}$$

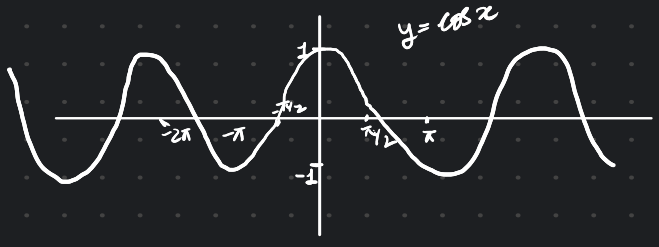
e.g. $f(x) = \dfrac{2x^4 - x^2 + 1}{x^2 - 4}$ Domain $= \mathbb{R} - \{2, -2\}$

Algebraic Functions:
made of algebraic operations (+, -, *, /) using other polynomials

Trigonometric functions



$\sin(n\pi) = 0$ n is an int.



$\cos(n\frac{\pi}{2}) = 0$, n is an int

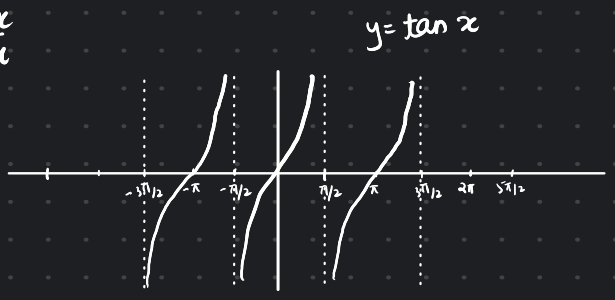
$\cos(n\pi) = 1$,

These graphs are periodic with a period 2π.

$\sin(x + 2\pi) = \sin x$

$\cos(x + 2\pi) = \cos x$

$\tan x = \frac{\sin x}{\cos x}$

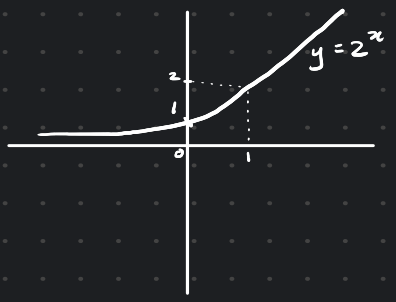


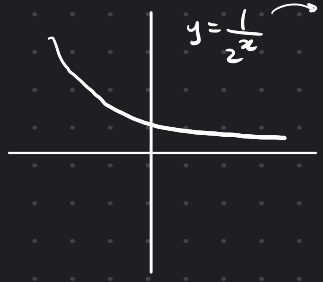
period = π

$\tan x = \tan(x + \pi)$

Exponential Function:

$f(x) = a^x$

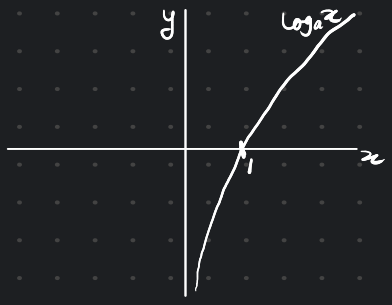




→ reciprocal of $y = 2^x$ (flip about the y-axis)

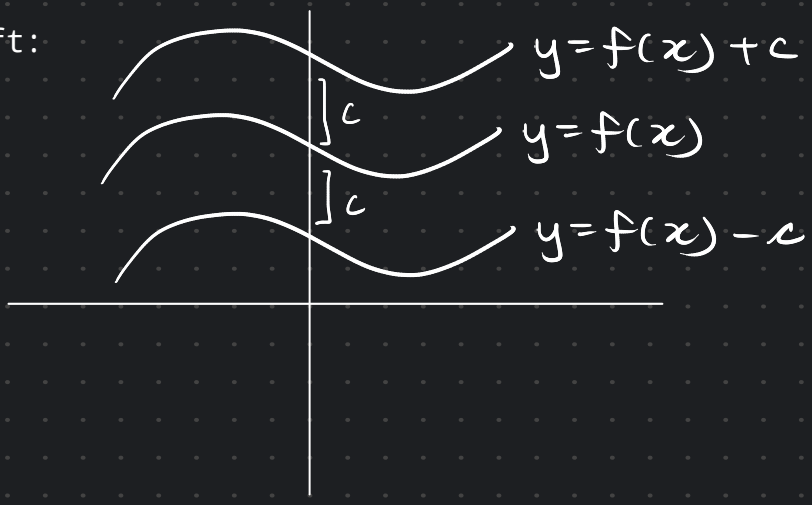
logarithmic function:

$f(x) = \log x$

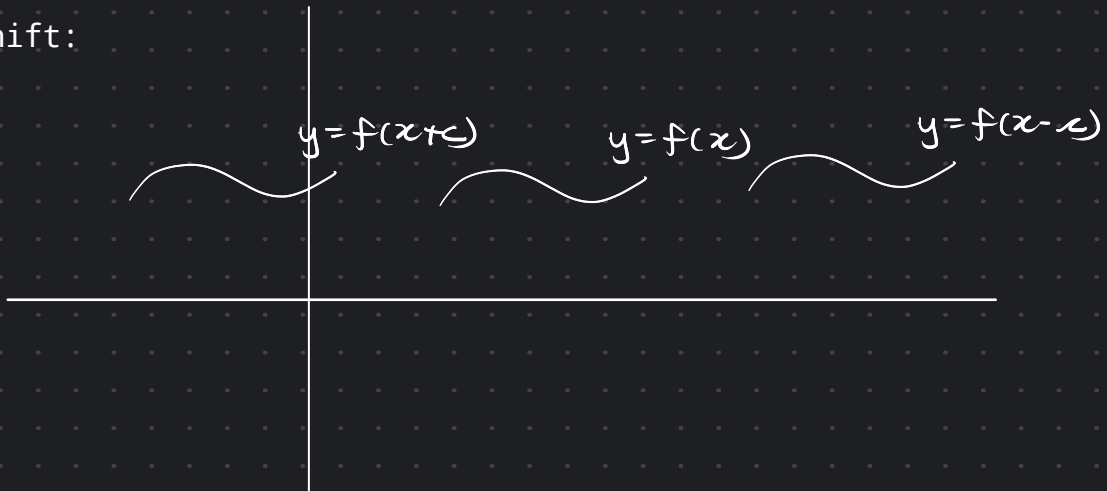


Transformation of functions:

vertical shift:

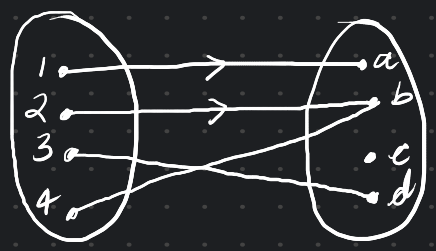


horizontal shift:



To be a function:
- every element of domain should be in codomain
- any element in the domain can have atmost 1 mapping in the co-domain

E.g. consider the relation R={(1, 2), (2, 3), (3, 4), (4, 5)}. It is a function.



Domain = {1, 2, 3, 4}
Range = {a, b, d}
Codomain = {a, b, c, d}

Q. Find domain and range for the following:

① $f(x) = \frac{1}{x-2}$

→ domain = $\mathbb{R} - \{2\}$
range = $\mathbb{R} - \{0\}$

② $f(x) = \frac{1}{x^2+2}$

$D = \mathbb{R}$
$R = (0, \frac{1}{2}]$

$y = \frac{1}{x^2+2}$ → $y > 0$
$x^2 + 2 = \frac{1}{y}$
$x^2 = \frac{1}{y} - 2 = \frac{1-2y}{y} = x^2$
$1 - 2y \geq 0$
$y \leq 1/2$
$y \neq 0$
$y \in (0, \frac{1}{2}]$

③ $f(x) = \frac{x^2-3x+2}{x^2-4x+3} = \frac{(x-1)(x-2)}{(x-1)(x-3)}$

Domain = $\mathbb{R} - \{1, 3\}$
Range = $\mathbb{R} - \{1, \frac{1}{2}\}$

$y = \frac{x-2}{x-3}$
$xy - 3y = x - 2$
$x(y-1) = 3y - 2$
$x = \frac{3y-2}{y-1}$
x cannot be 1 or 3
$x \neq 1$: $3y - 2 \neq y - 1$, $2y \neq 1$, $y \neq 1/2$
$x \neq 3$: $3y - 3 = 3y - 2$ ✗

Piece-wise functions:

e.g. $f(x) = \begin{cases} x-2, & x \geq 2 \\ 2-x, & x < 2 \end{cases}$ ⇒ $f(x) = |x-2|$

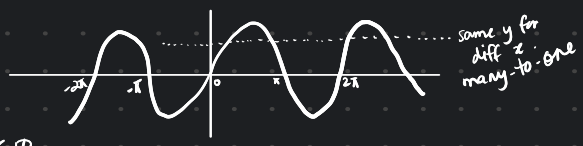
Identical functions:
Two functions f and g are identical if:
- Domain(f) = Domain(g)
- Range(f) = Range(g)
- f(x) = g(x) for all x ∈ D

Types of mappings:

i. one-to-one:
- each element in the domain of f has a distinct image in the co-domain.

ii. many-to-one:
- there are atleast 2 elements in the domain whose images are the same.

e.g. $f(x) = \sin x$



e.g. $f(x) = x^3 + x + 1, \quad x \in \mathbb{R}$

→ Assume it is one-one → Then $x_1 = x_2$ [If $x_1 \neq x_2$, we'd conclude otherwise]

$f(x_1) = f(x_2)$

$x_1^3 + x_1 + 1 = x_2^3 + x_2 + 1$

$x_1^3 - x_2^3 + x_1 - x_2 = 0$

$(x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) + (x_1 - x_2) = 0$

$\Rightarrow (x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2 + 1) = 0$

$x_1 - x_2 = 0$ or $x_1^2 + x_1x_2 + x_2^2 + 1 = 0$

$x_1 = x_2$ → one-one

$x_1 = \frac{-x_2 \pm \sqrt{x_2^2 - 4(x_2^2+1)}}{2}$

$= \frac{-x_2 \pm \sqrt{-3x_2^2 - 4}}{2}$

$= -\frac{x_2 \pm i\sqrt{3x^2+4}}{2}$ → ×

∴ It is a one-one function.

Onto and into functions:

A function is onto if each element in the codomain has atleast 1 pre-image in the domain.

$f: X \to Y, \ \forall y \in Y \ \exists x \in X \rightarrow$ onto

i.e. Onto if range == co-domain.

A function is into if there's something in the co-domain which nothing in the domain is mapped to.

$f: X \to Y; \ \forall y \in Y \ \nexists x \in X \rightarrow$ into

i.e. range $\subset$ co-domain

A function is bijective if it is both one-one and onto.

surjective :: onto
injective :: one-one
bijective :: injective and surjective

To check if a function is onto:
- find range and codomain, check if range == codomain or
- solve f(x) = y for x
  say x = g(y)
  if g(y) is defined for each y ∈ Co-domain,
  and g(y) ∈ domain for all y ∈ co-domain,
  f(x) is onto.

Q. $f(x): \mathbb{R} \to \mathbb{R}$ $f(x) = \sin x$

→ Range = $[-1, 1]$ ∴ Range $\subset$ co-domain [there is stuff in domain with no pre-image]
∴ $f(x)$ is into.

To be onto,
$f(x): \mathbb{R} \to [-1, 1] \Rightarrow \sin x$ will be onto.

Q. $f: \mathbb{R} \to A$, $f(x) = \frac{x^2}{x^2+1}$ is surjective. Find A

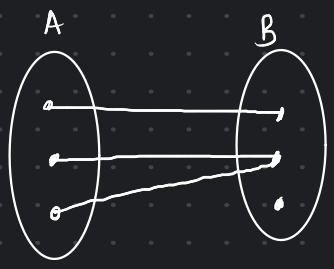
→ Given, $f(x)$ is onto. ∴ Range == co-domain

$y = \frac{x^2}{x^2+1}$
$x^2(y-1) = -y$
$x^2 = \frac{y}{1-y}$ $y \geq 0$
$y \neq 1, y \leq 1$

∴ Range = $[0, 1)$
$\Rightarrow A = [0, 1)$ (surjective)

Functions of a single variable

A function f from a set A to a set B is a rule that assigns to each element x in A, one element in B.



A, the values that x takes is the domain. $D = \{x \mid x \in A\}$

The values that f(x) takes is the range. It is a subset of the codomain, B. $R \subseteq B$

Q. Domain of $f(x) = \sqrt{x^2 - 9}$?

→ $x^2 - 9 \geq 0$

$x^2 \geq 9$

$x \in (-\infty, -3] \cup [3, \infty)$

Some standard functions

Linear functions:

Properties of functions

1. Let f(x) and g(x) be functions with domains A and B.
   (f+g)(x) = f(x) + g(x) with domain A ∩ B

2. fog(x) = f(g(x))
   gof(x) = g(f(x))

If $g(x) = 1 - x$ and $h(x) = \frac{x}{x-1}$, then $\frac{g(h(x))}{h(g(x))}$ is:

A. $\frac{h(x)}{g(x)}$
B. $\frac{-1}{x}$
C. $\frac{g(x)}{h(x)}$
D. $\frac{x}{(1-x)^2}$

→ $\frac{g(h(x))}{h(g(x))} = \frac{1 - \frac{x}{x-1}}{\frac{1-x}{1-x-1}} = \frac{-1}{x-1} \times \frac{-x}{1-x} = \frac{-x}{(1-x)^2} = \frac{x}{(x-1)} \cdot \frac{1}{(1-x)} = \frac{h(x)}{g(x)}$

$$f(x) = \log\left(x + \sqrt{1+x^2}\right)$$

$$f(-x) = \log\left(-x + \sqrt{1+x^2}\right)$$

$$f(x) + f(-x) = \log\left(x + \sqrt{1+x^2}\right) + \log\left(-x + \sqrt{1+x^2}\right)$$

$$= \log\left[\left(x + \sqrt{1+x^2}\right)\left(-x + \sqrt{1+x^2}\right)\right]$$

$$= \log\left[1 + x - x^2\right]$$

$$= 0$$

$$f(x) = -f(-x) \Rightarrow \text{odd}.$$

Inverse functions

$f(x) = 2(x+100)$

$f'(x) = \frac{x}{2} - 100$

$f'(2(x+100)) = \frac{2(x+100)}{2} - 100 = x$ $\quad f'(f(x)) = x$ $\quad$ Ily, $f(f'(x)) = x$

Q. $f(x) = 3x - 4$

→ $y = 3x - 4$

$x = \frac{y+4}{3}$

$f'(x) = \frac{x+4}{3}$

Q. $f(x) = \log_a(x + \sqrt{x^2+1})$; $a > 0$, $a \neq 1$

f ↓ y

$a^y = x + \sqrt{x^2+1}$

$(a^y - x)^2 = (\sqrt{x^2+1})^2$

$a^{2y} - 2xa^y + x^2 = x^2 + 1$

$x = \frac{a^{2y} - 1}{2a^y}$

Q. Find inverse of $f(x) = x^2$. $x \geq 0$

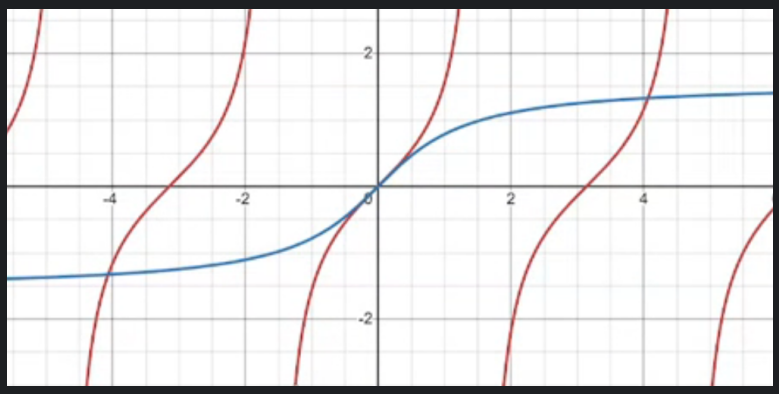
→ $\sqrt[2]{x}$

To check if inverse for $f(x)$ exists: horizontal line test.

	Domain	Range
$\tan x$	$\mathbb{R} - (2n+1)\frac{\pi}{2}$	$(-\infty, \infty)$
$\tan^{-1} x$	$(-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot x$	$\mathbb{R} - n\pi$	$(-\infty, \infty)$
$\cot^{-1} x$	$(-\infty, \infty)$	$(0, \pi)$

$\tan(\frac{\pi}{2}) = \infty$
$\Rightarrow \tan^{-1}(\infty) = \frac{\pi}{2}$
Ily $\tan^{-1}(-\infty) = -\frac{\pi}{2}$

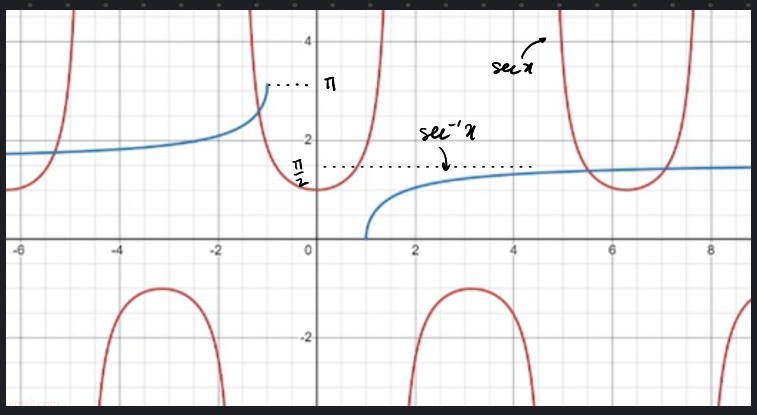
$\tan x$



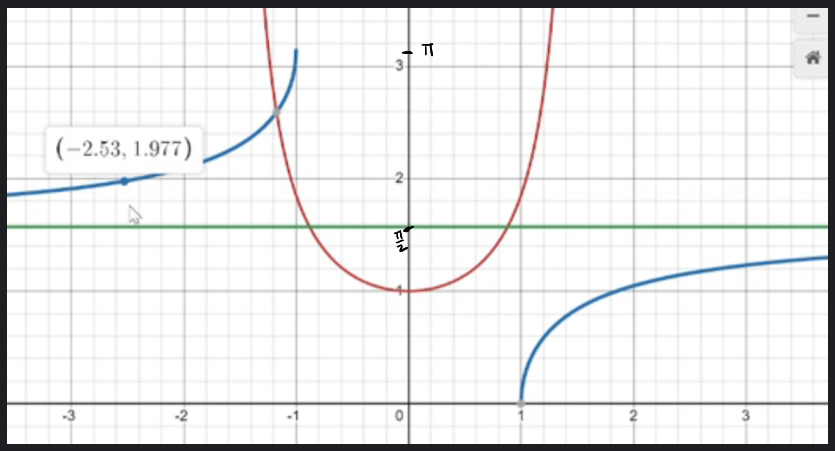
$\tan^{-1} x$

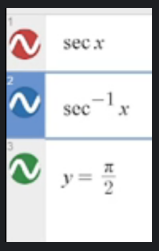
	Domain	Range
$\sec x$	$\mathbb{R} - (2n+1)\frac{\pi}{2}$	$(-\infty, -1] \cup [1, \infty)$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$
$\operatorname{cosec} x$	$\mathbb{R} - n\pi$	$(-\infty, -1] \cup [1, \infty)$
$\operatorname{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$

$\sec x$



$\sec^{-1} x$





If $f(x)$ & $g(x)$ → inverse of $f(x)$

~~(b)~~ $f(x)$ & $g(x)$ must be strictly increasing function / strictly decreasing function

True or False

true